

PROJECT PHOENIX

Architectural Master Specification v1.0

Executable Specification

Status: Executable Specification - Human + Machine + Python Contract

Application: BREWSTER ENGINEERING WIZARD. Platform: PROJECT PHOENIX.

1. Executive Purpose

This document defines the executable specification for the Project Phoenix Architectural Suite. It is the single source of truth for implementation, validation, documentation and future code generation. The specification is designed to be consumed by humans, by Phoenix Intelligence, and by Python validators.

2. Architecture Principles

- specification_first
- single_source_of_truth
- pkb_bib_integrated
- digital_twin_synchronization
- traceable_outputs

3. Platform Context

The Architectural Suite runs inside Project Phoenix. Brewster Engineering Wizard is the application shell and user-facing product. Phoenix Core provides runtime, update, registry and health services. Phoenix Intelligence provides PKB, BIB, Knowledge Graph, Decision Log and Source Evidence.

4. Executable Specification Model

Every architectural module must define: id, name, purpose, inputs, outputs, knowledge_sources and tests. A module cannot be considered implementation-ready without these fields.

5. Module Catalog

ID	Name	Purpose	Inputs	Outputs	Tests
architectural.project_intake	Project Intake	Collects project goals, location, scope, constraints and deliverables.	user_brief, location, project_type	project_profile.json, intake_report.md	test_required_fields, test_output_profile
architectural.space_program	Programma van Eisen / Space Program	Transforms user requirements into rooms, areas, functions and relationships.	project_profile, reference_templates, regulations	space_program.json, room_schedule.xlsx	test_area_balance, test_room_schema
architectural.room_schedule	Ruimtestaat	Creates a structured room schedule with function, area, floor, occupancy and notes.	space_program	room_schedule.json, room_schedule.csv, room_schedule.xlsx	test_room_ids, test_area_totals

architectural.floorplan_generator	Plattegrondgenerator	Generates schematic and later CAD-grade floorplans.	project_profile, space_program, parcel, constraints	floorplan.json, floorplan.dxf, floorplan.pdf	test_room_generation, test_dimensioning, test_accessibility
architectural.facade_generator	Gevelgenerator	Generates facade concepts, openings, materials and facade drawings.	floorplan, levels, openings, materials	facades.json, facades.dxf, facades.pdf	test_opening_alignment, test_levels
architectural.section_generator	Doorsnedegenerator	Generates building sections, storey heights, structural zones and roof build-ups.	floorplan, levels, roof, structure	sections.json, sections.dxf, sections.pdf	test_section_levels, test_headroom
architectural.roof_generator	Dakgenerator	Generates roof geometry, slopes, drainage zones and build-up.	building_mass, climate, drainage_rules	roof.json, roof_plan.dxf, roof_plan.pdf	test_roof_geometry, test_drainage
architectural.openings	Deuren en kozijnen	Places doors/windows and manages schedules.	floorplan, facades, room_schedule	opening_schedule.xlsx, openings.json	test_door_widths, test_window_room_links
architectural.stairs	Trapgenerator	Creates stairs, checks basic rise/run and circulation.	levels, floorplan, accessibility_rules	stair.json, stair_details.pdf	test_rise_run, test_clear_width
architectural.materials	Materialisatie	Assigns materials to walls, floors, roofs, facades and interior finishes.	building_elements, pkb_material_library	material_schedule.xlsx, materials.json	test_material_ids, test_element_materials
architectural.quantity_takeoff	Hoeveelhedenstaat	Calculates quantities for areas, lengths, openings and elements.	digital_twin, building_elements	quantity_takeoff.xlsx, quantity_takeoff.json	test_quantity_units, test_quantity_totals
architectural.compliance_precheck	Bouwregelcontrole pre-check	Performs pre-checks for building rules, accessibility, fire safety, daylight and ventilation.	project_profile, floorplan, room_schedule, regulations	compliance_precheck.md, issues.json	test_rule_references, test_issue_schema
architectural.bim_export	BIM Export	Exports architectural objects to BIM/Digital Twin payloads and IFC-ready structures.	digital_twin, building_elements	architectural_twin.json, ifc_payload.json	test_object_ids, test_ifc_mapping
architectural.drawing_package	Vergunningstekeningpakket	Composes the drawing package for PDF/DXF/DWG/SKP/IFC outputs.	floorplans, facades, sections, schedules	drawing_package.pdf, drawing_index.json	test_sheet_index, test_required_drawings
architectural.report_export	Rapportage export	Exports DOCX/PDF reports and specification summaries.	all_module_results	architectural_report.docx, architectural_report.pdf	test_report_sections, test_export_exists

6. Data and Knowledge Contracts

The suite reads from PKB, BIB, Knowledge Graph, Decision Log, Source Evidence, standards, regulations, templates and project data. All important outputs must be traceable through STEE/source evidence.

7. Workflow Contract

1. Project Intake
2. Programma van Eisen
3. Room Schedule
4. Concept Variants
5. Floorplans

6. Facades
7. Sections
8. Roof
9. Openings
10. Materials
11. Quantity Takeoff
12. Compliance Pre-check
13. BIM/Digital Twin
14. Drawing Package
15. Report Export

8. Validation Gates

- schema_valid
- tests_defined
- outputs_declared
- knowledge_sources_declared
- stree_traceability_declared

9. Digital Twin Synchronization

Every geometry-generating module must write deterministic object identifiers, classification, metadata, source references and version stamps to the Digital Twin payload.

10. Implementation Roadmap

v1.0 creates the executable specification and validation contract. v1.1 creates module scaffolding. v1.2 implements Project Intake, Space Program and Room Schedule. v1.3 implements Floorplan Generator. v1.4 adds facade, section and roof generators. v1.5 adds export, compliance and Digital Twin synchronization.

Module Contract: architectural.project_intake

Name: Project Intake

Purpose: Collects project goals, location, scope, constraints and deliverables.

Inputs: user_brief, location, project_type

Outputs: project_profile.json, intake_report.md

Knowledge sources: pkb, bib, decision_log

Required tests: test_required_fields, test_output_profile

Module Contract: architectural.space_program

Name: Programma van Eisen / Space Program

Purpose: Transforms user requirements into rooms, areas, functions and relationships.

Inputs: project_profile, reference_templates, regulations

Outputs: space_program.json, room_schedule.xlsx

Knowledge sources: pkb, bib, templates

Required tests: test_area_balance, test_room_schema

Module Contract: architectural.room_schedule

Name: Ruimtestaat

Purpose: Creates a structured room schedule with function, area, floor, occupancy and notes.

Inputs: space_program

Outputs: room_schedule.json, room_schedule.csv, room_schedule.xlsx

Knowledge sources: pkb, standards

Required tests: test_room_ids, test_area_totals

Module Contract: architectural.floorplan_generator

Name: Plattegrondgenerator

Purpose: Generates schematic and later CAD-grade floorplans.

Inputs: project_profile, space_program, parcel, constraints

Outputs: floorplan.json, floorplan.dxf, floorplan.pdf

Knowledge sources: pkb, knowledge_graph, digital_twin

Required tests: test_room_generation, test_dimensioning, test_accessibility

Module Contract: architectural.facade_generator

Name: Gevelgenerator

Purpose: Generates facade concepts, openings, materials and facade drawings.

Inputs: floorplan, levels, openings, materials

Outputs: facades.json, facades.dxf, facades.pdf

Knowledge sources: pkb, templates

Required tests: test_opening_alignment, test_levels

Module Contract: architectural.section_generator

Name: Doorsnedegenerator

Purpose: Generates building sections, storey heights, structural zones and roof build-ups.

Inputs: floorplan, levels, roof, structure

Outputs: sections.json, sections.dxf, sections.pdf

Knowledge sources: pkb, digital_twin

Required tests: test_section_levels, test_headroom

Module Contract: architectural.roof_generator

Name: Dakgenerator

Purpose: Generates roof geometry, slopes, drainage zones and build-up.

Inputs: building_mass, climate, drainage_rules

Outputs: roof.json, roof_plan.dxf, roof_plan.pdf

Knowledge sources: standards, pkb

Required tests: test_roof_geometry, test_drainage

Module Contract: architectural.openings

Name: Deuren en kozijnen

Purpose: Places doors/windows and manages schedules.

Inputs: floorplan, facades, room_schedule

Outputs: opening_schedule.xlsx, openings.json

Knowledge sources: templates, standards

Required tests: test_door_widths, test_window_room_links

Module Contract: architectural.stairs

Name: Trapgenerator

Purpose: Creates stairs, checks basic rise/run and circulation.

Inputs: levels, floorplan, accessibility_rules

Outputs: stair.json, stair_details.pdf

Knowledge sources: standards, pkb

Required tests: test_rise_run, test_clear_width

Module Contract: architectural.materials

Name: Materialisatie

Purpose: Assigns materials to walls, floors, roofs, facades and interior finishes.

Inputs: building_elements, pkb_material_library

Outputs: material_schedule.xlsx, materials.json

Knowledge sources: pkb, bib

Required tests: test_material_ids, test_element_materials

Module Contract: architectural.quantity_takeoff

Name: Hoeveelhedenstaat

Purpose: Calculates quantities for areas, lengths, openings and elements.

Inputs: digital_twin, building_elements

Outputs: quantity_takeoff.xlsx, quantity_takeoff.json

Knowledge sources: digital_twin

Required tests: test_quantity_units, test_quantity_totals

Module Contract: architectural.compliance_precheck

Name: Bouwregelcontrole pre-check

Purpose: Performs pre-checks for building rules, accessibility, fire safety, daylight and ventilation.

Inputs: project_profile, floorplan, room_schedule, regulations

Outputs: compliance_precheck.md, issues.json

Knowledge sources: pkb, standards, source_evidence

Required tests: test_rule_references, test_issue_schema

Module Contract: architectural.bim_export

Name: BIM Export

Purpose: Exports architectural objects to BIM/Digital Twin payloads and IFC-ready structures.

Inputs: digital_twin, building_elements

Outputs: architectural_twin.json, ifc_payload.json

Knowledge sources: digital_twin, knowledge_graph

Required tests: test_object_ids, test_ifc_mapping

Module Contract: architectural.drawing_package

Name: Vergunningstekeningpakket

Purpose: Composes the drawing package for PDF/DXF/DWG/SKP/IFC outputs.

Inputs: floorplans, facades, sections, schedules

Outputs: drawing_package.pdf, drawing_index.json

Knowledge sources: templates, source_evidence

Required tests: test_sheet_index, test_required_drawings

Module Contract: architectural.report_export

Name: Rapportage export

Purpose: Exports DOCX/PDF reports and specification summaries.

Inputs: all_module_results

Outputs: architectural_report.docx, architectural_report.pdf

Knowledge sources: pkb, templates

Required tests: test_report_sections, test_export_exists